SUPPORT VECTOR MACHINE (SVM) – COMPLETE ASSIGNMENT

Dataset: Mushroom Dataset

1. INTRODUCTION

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It identifies the optimal hyperplane that maximizes the margin between different classes. In this assignment, we classify mushrooms as edible (e) or poisonous (p).

2. EXPLORATORY DATA ANALYSIS (EDA)

- The dataset consists entirely of categorical features.
- No missing values are present.
- The dataset is balanced between edible and poisonous classes.
- Key observations:
    - **Odor** is the most important feature.
    - Features like **gill size, cap color, and habitat** also influence classification.

Visualizations used:

- Count plots
- Distribution plots

3. DATA PREPROCESSING

- All categorical features were encoded using **Label Encoding**.
- Dataset split:
    - **80% Training**
    - **20% Testing**

- Feature scaling applied using **StandardScaler** (important for SVM).

4. DATA VISUALIZATION

- Pairplots used to understand feature relationships.
- Count plots used to analyze class distribution.
- Heatmap used for correlation analysis.

5. SVM IMPLEMENTATION

- Model used: **Support Vector Classifier (SVC)**
- Trained using training data.
- Evaluated using:
    - Accuracy
    - Precision
    - Recall
    - F1-score

6. VISUALIZATION OF RESULTS

- Confusion Matrix plotted.
- Classification report generated.
- Clear separation observed between classes.

7. PARAMETER TUNING

- Used **GridSearchCV** to optimize:
    - Kernel: Linear, Polynomial, RBF
    - C (Regularization parameter)

8. KERNEL COMPARISON

Kernel	Performance

Linear	Best (fast & accurate)
Polynomial	Moderate
RBF	High accuracy but slower

9. STRENGTHS & WEAKNESSES

Strengths:

- Works well in high-dimensional data
- Effective for binary classification
- Robust against overfitting

Weaknesses:

- Computationally expensive
- Requires tuning
- Less interpretable

10. CONCLUSION

SVM performed extremely well on the Mushroom dataset due to its near-linear separability. The **Linear kernel** provided the best balance between speed and accuracy.